Chapter 14

Microbial Genetics

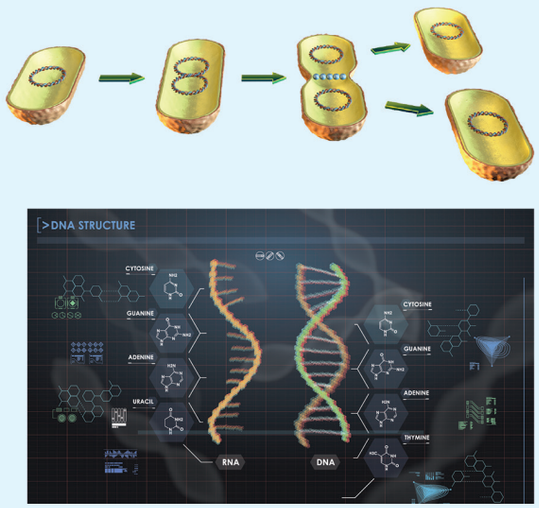
Chapter Outline

14.1 Genetic Information is Stored in DNA

14.2 Structure of DNA

14.3 DNA Replication

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Microbial genetics provides powerful tools for deciphering the regulation, as well as the functional and pathway organization of cellular processes. The genetic study of microbes has played a highly significant role in the developments of Molecular Biology, Recombinant DNA Technology and in the preparation of useful products. Microbial Genetics makes microbes beautiful, beneficial and bountiful.

Learning Objectives

After studying this chapter the student will be able,

- To *review the historical discoveries that led to establishing DNA as the genetic material.*
- *To identify the role of genetic material.*
- *To recognize the contributions of Griffith, Avery, MacLeod, and McCarty, and Hershey and Chase.*
- *To explain the structure of DNA.*
- *To recognize the contributions of Chargaff, Rosalind Franklin, Maurice Wilkins, Watson and Crick.*
- *To describe the Watson and Crick model of DNA.*
- *To compare structure of DNA and RNA.*
- *To know Meselson and Stahl's experiment.*
- *To explain the steps of replication.*
- *To know the enzymes and their roles involved in DNA replication.*

14.1 Genetic Information is Stored in DNA

Microorganisms are diverse in nature. A particular bacterium can be identified based on certain characteristics. When a bacterial cell grows and divides, it gives rise to cells with similar characteristics. Have you ever pondered as to why some of the characteristics of progeny cells are similar and a few dissimilar?

In the middle of the 19th century it was assumed that there was some particle present somewhere in the cell which